

JEE MAIN REVISION MASTER: DIFFERENTIATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 83 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Evaluation of Limits & L'Hopital's Rule	57 PYQs	Limits
Very High	Continuity & Differentiability at a Point	26 PYQs	Continuity & Differentiability

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Evaluation of Limits & L'Hopital's Rule	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$, $\lim_{x \rightarrow a} f(x)^{g(x)} = e^{\lim_{x \rightarrow a} (f(x) - 1)g(x)}$ (for 1^∞)
Continuity & Differentiability at a Point	$\text{Continuous at } a \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$, $\text{Differentiability} \implies \text{Continuity}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Evaluation of Limits & L'Hopital's Rule (Very High Yield — 57 PYQs)

Core Principle: Resolution of indeterminate forms ($0/0$, ∞/∞ , 1^∞), standard trigonometric and exponential limits, and Series expansion.

■ **Caution / JEE Pitfall:** Before applying L'Hopital's rule, ALWAYS check that the limit is in an indeterminate form ($0/0$ or ∞/∞). Differentiating a determinate form produces an incorrect answer.

■ **Speed Trick:** Maclaurin expansions are often faster than repeated L'Hopital: $\sin x = x - (x^3/6)$, $\cos x = 1 - (x^2/2)$, $e^x = 1 + x + (x^2/2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Let $f(x)=x^3+x^2f'(1)+2xf''(2)+f'''(3)$, $x \in \mathbb{R}$. Then the value of $f'(5)$ is :..." [Ans: N/A]
- [2024 Morning Shift] Q#1 (MCQ): "Suppose $f(x)=\frac{2^x+2^{-x}}{\tan x} \sqrt{(\tan^2-1)(x^2-x+1)(7x^2+3x+1)^3}$. Then the value of $f'(0)$ is equal to..." [Ans: N/A]
- [2024 Evening Shift] Q#2 (MCQ): "If $\log_e y=3\sin^{-1}x$, then $(1-x^2)y^{n-xy'}$ at $x=(1/2)$ is equal to..." [Ans: N/A]

3.2 Continuity & Differentiability at a Point (Very High Yield — 26 PYQs)

Core Principle: Epsilon-delta and limit criteria for continuity, left/right hand derivatives, and graphical identification of non-differentiable sharp points.

■ **Caution / JEE Pitfall:** Differentiability implies continuity, but continuity DOES NOT imply differentiability (e.g. $f(x) = |x|$ is continuous at $x=0$ but not differentiable).

■ **Speed Trick:** The greatest integer function $[x]$ is discontinuous at all integer points $x \in \mathbb{Z}$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$, then (dy/dx) at $x = \frac{\sqrt{3}}{2}$ is e..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#1 (MCQ):** "Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a function which is differentiable at all points of its domain and satisfies the condition $x^2 f'(x) = f(x)$..." **[Ans: N/A]**
- **[2025 Morning Shift] Q#1 (Numerical):** "Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable odd function satisfying $f''(x) \geq 0$, $f''(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____." **[Ans: 36]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 2nd April Morning Shift (Q#1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable odd function satisfying $f''(x) \geq 0$, $f''(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____.

Correct Answer: 36

Solution Analysis: $f''(x) \geq 0$, $f''(x) = f(x)$ Second order differential equation $f(x) = Ae^x + Be^{-x}$ $f(0) = 0 \Rightarrow A = -B \Rightarrow f(x) = A(e^x - e^{-x})$ $f'(x) = A(e^x + e^{-x}) = A(e^x + e^{-x})$ $f'(0) = 3 = A(e^0 + e^0) = 2A \Rightarrow A = (3/2)$ $f(x) = (3/2)(e^x - e^{-x})$ If $\ln 3 = (27/2)(e^{\ln 3} - e^{-\ln 3}) = (27/2)(3 - (1/3)) = (27/2) \cdot (8/3) = 36$

Benchmark Problem 2: JEE Main 2024 (Online) 1st February Evening Shift (Q#3)

If $y = \frac{\sqrt{x}+1}{x^2-\sqrt{x}} \cdot x\sqrt{x} + x + \sqrt{x} + (1/15)(3\cos^2 x - 5)\cos^3 x$, then $96y'(\pi/6)$ is equal to :

Correct Answer: 105

Solution Analysis: $y = \frac{\sqrt{x}+1}{x^2-\sqrt{x}} \cdot x\sqrt{x} + x + \sqrt{x} + (1/15)(3\cos^2 x - 5)\cos^3 x$
 $y = \frac{\sqrt{x}+1}{(\sqrt{x})^3-1} \cdot (\sqrt{x})((\sqrt{x})^2+(\sqrt{x})+1) + (1/15)\cos^5 x - (1/3)\cos^3 x$
 $y = (\sqrt{x}+1)(\sqrt{x}-1) + (1/15)\cos^5 x - (1/3)\cos^3 x$
 $y' = 1 - \cos^4 x + \cos^2 x \sin x + \cos^2 x \sin x$
 $y'(\pi/6) = 1 - \frac{1}{16} + \frac{3}{4} = \frac{32-9+12}{32} = \frac{35}{32}$
 $\therefore 96y'(\pi/6) = 105$